

LEARNER WORKBOOK

FULL NAME & SURNAME	Zwelethu Menyuka
ID NUMBER:	8610295538084
NAME OF ASSESSOR	COURTNEY ROSENBERG
DATE OF ASSESSMENT	22.06.2021
VENUE	Pretoria Venue: 2nd Floor, 679 Paul Kruger Street

	ACHIEVED	NOT ACHIEVED	<i>Well Done</i>
KNOWLEDGE	☒	☐	
SKILLS	☐	☐	



Signature of learner



Signature of Assessor

COURTNEY ROSENBERG

Learner workbook





In terms of the Copyright Act 1978:

No part of this manual may be reproduced or translated in any form or by any means, electronic or mechanical including photocopying, recording or by an information storage and retrieval system without the express permission from TDA.

COPYRIGHT: TRAINING DYNAMICS AFRICA

First edition 2007

Publications disclaimer

Neither the authors nor the SETA can accept any responsibility for any legal or any other consequences, which may arise directly or indirectly as a result of the use or adaptation of or reliance upon any of the content of this material.

This material has been compiled based on the **UNIT STANDARD 9013 DESCRIBE, APPLY, ANALYSE AND CALCULATE SHAPE AND MOTION IN 2-AND 3-DIMENSIONAL SPACE IN DIFFERENT CONTEXT.**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by: Training Dynamics Africa

Learner workbook





HOW THIS LEARNER WORKBOOK IS COMPILED

This learner workbook has been compiled to provide learners towards the mathematical literacy requirements of the NQF at Level 3.

People credited with this unit standard are able to:

1. Measure, estimate, and calculate physical quantities in practical situations relevant to the adult in life or the workplace.
2. Explore describe and represent, interpret and justify geometrical relationships and conjectures to solve problems in two and three dimensional geometrical situations.

You have to study the content and guidelines given in your learner reference guide to complete this workbook.

Remember that this completed workbook must be handed in as it will form part of your formative assessment and will play a role in awarding you the required 04 credits that are to be achieved upon completion of this module.

Write neatly and legibly and ensure that you do not simply copy the information contained in the learner reference guide into your workbook. Give your own opinion and ideas where applicable and write the answers in your own words to indicate your level of understanding and learning.

Good luck with the completion of this workbook.

LEARNING UNIT 1

Metric System

LEARNING OUTCOMES

A learning credited with this unit will be able to:

1. Explain what is meant with accuracy.
2. Explain the term rounding.
3. Do basic calculations using: addition, subtraction, multiplication, division, powers and roots.
4. Apply the measurement system and instruments used in South Africa as set out in this guide.
5. Apply area and volume calculation within an operational environment.
6. Compile a rough sketch of an area or scene of incident.
7. Interpret a plan drawn according to scale.



LEARNER ACTIVITY 1

Convert the following using the metric system:

a) 35 m = $\frac{35\text{m} \times 1000 = 35\,000}{}$ mm

b) 0,7 m = $\frac{0,7\text{m} \times 100 = 70}{}$ cm

c) 56 mm = $\frac{56\text{mm} / 1000 = 0,056}{}$ m

d) 80 km = $\frac{80\text{km} \times 1000 = 80\,000}{}$ m

e) 32 cm = $\frac{32\text{cm} \times 10 = 320}{}$ mm

f) 790 cm = $\frac{790\text{cm} / 100 = 7,9}{}$ m

g) 38 543 g = $\frac{38\,543\text{g} / 1000 = 38,543}{}$ kg

h) 789 mg = $\frac{789\text{mg} / 10 = 78,9}{}$ g

i) 2,5 l = $\frac{2,5\text{l} \times 1000 = 2500}{}$ ml

j) 90 ml = $\frac{90\text{ml} / 1000 = 0,09}{}$ l





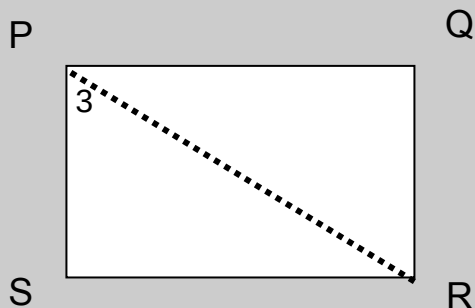
Requirements achieved	YES	☒	☐	NO	☐
Comments in support of decision					

Achieved

LEARNER ACTIVITY 2

Use the theorem of Pythagoras to calculate the following:

- ♦ A rectangular field is to be divided along its diagonal PR into two plots. If the dimensions of the field are 150m × 75m calculate the length of the fencing needed to divide the field (length of PR)
- ♦ Round off to an acceptable level of accuracy.



$$PR^2 = PQ^2 + QR^2$$

$$= (150)^2 + (75)^2$$

$$PR^2 = 28125$$

$$PR = \sqrt{28125}$$

$$PR = 167,705$$



Requirements achieved

YES



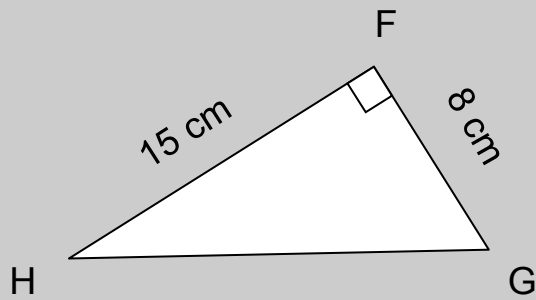
NO

Comments in support of decision

Achieved

LEARNER ACTIVITY 3

Calculate the length of side GH in $\triangle FGH$.



$$CD = DE + CE$$

$$= (13\text{cm}) + (12\text{cm})$$

$$CD = 25\text{cm}$$

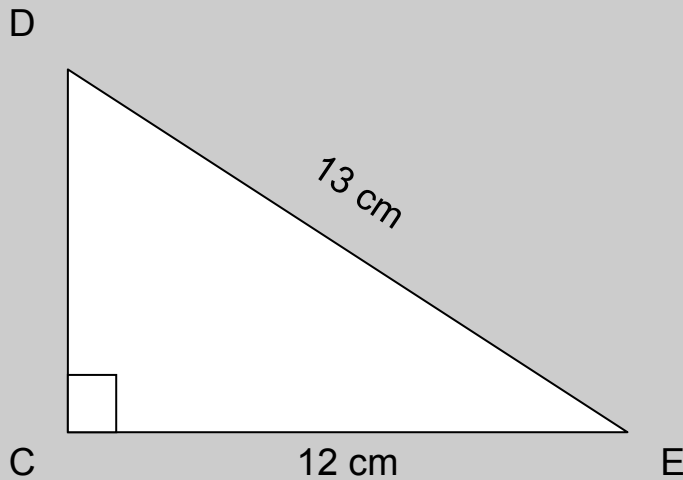
$$CD = 5\text{cm}$$

Requirements achieved	YES	☒	☐	NO
-----------------------	-----	-------------------------------------	--------------------------	----

Comments in support of decision

Achieved

LEARNER ACTIVITY 4

Find the length of side CD in $\triangle CDE$.

$$CD = DE + CE$$

$$= (13\text{cm}) + (12\text{cm})$$

$$CD = 25\text{cm}$$

$$CD = 5\text{cm}$$



Requirements achieved

YES



NO

Comments in support of decision

Achieved

LEARNER ACTIVITY 5

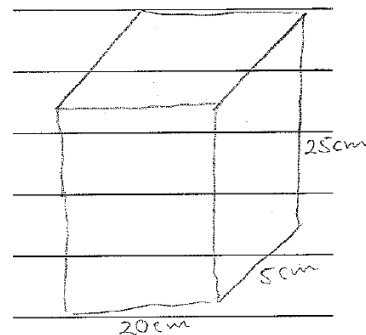
Use the formulae of perimeter, area, volume and surface area to calculate the following:

- ♦ Find the volume of a box with a rectangular base measuring 20 cm by 5 cm and a height of 25 cm. (Draw a rough sketch of the box).
- ii) How many of these boxes can be packed into a crate measuring 50 cm by 35 cm by 40 cm?
- iii) Find the surface area of the box (draw a net of it first).

i) Volume = $l \times b \times h$

$$= (20\text{cm}) (5\text{cm}) (25\text{cm})$$

$$= 2500\text{cm}^3$$



ii) Volume of a crate = $l \times b \times h$

$$= (50\text{cm}) (35\text{cm}) (40\text{cm})$$

$$= 70\,000\text{cm}^3$$



$$\text{Volume of crate} / \text{volume of box} = 70\,000\text{cm}^3 / 2500\text{cm}^3$$

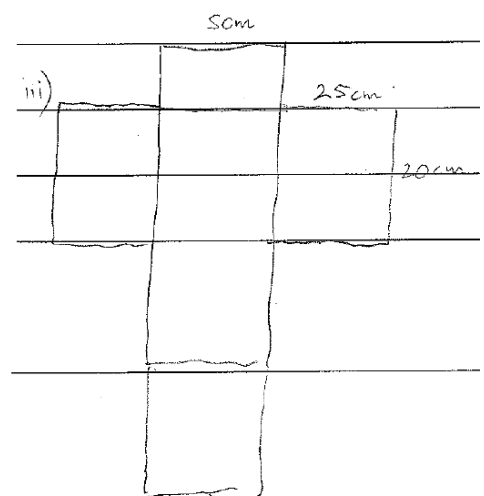
$$= 28 \text{ boxes}$$

iii) Surface Area = $(l \times b) \times 2 + (h \times b) \times 2 + (l \times h) \times 2$

$$= (20 \times 5) \times 2 + (25 \times 5) \times 2 + (20 \times 25) \times 2$$

$$= 200 + 250 + 1000$$

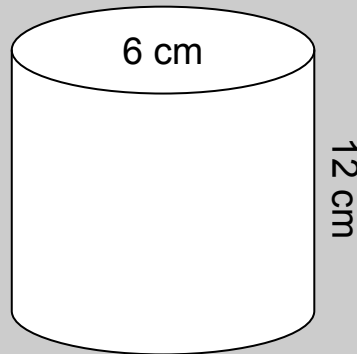
$$= 1450\text{cm}^2$$



LEARNER ACTIVITY 6

A cylindrical can has a height of 12 cm and a diameter of 6 cm. Find:

- its volume
- its surface area (draw a net of it first)



- Radius = $\frac{1}{2}$ diameter

$$\text{Volume} = \pi r^2 h$$

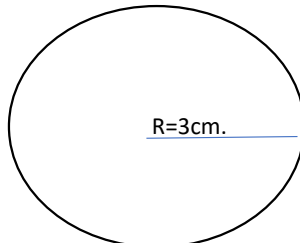
$$V = 3.14 \times 3^2 \times 12$$

$$V = 3.14 \times 9 \times 12$$

$$V = 339.12 \text{ cm}^2$$



-



$$\text{Total Surface Area} = 2\pi r^2 + 2\pi rh$$

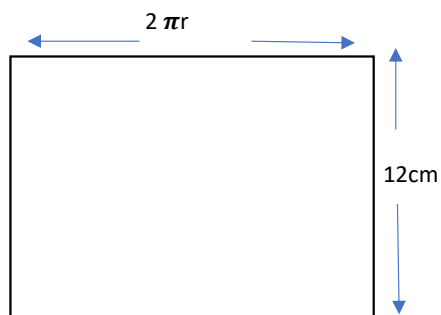
$$= 2(\pi)(3\text{cm})^2 + 2(\pi)(3\text{cm})(12\text{cm})$$

$$= 56,548.668\text{cm}^2 + 226.194.67\text{cm}^2$$

$$= 282.74\text{cm}^2$$

$$= 283\text{cm}^2$$

-



Requirements achieved	YES	☒	☐	NO	☐
Comments in support of decision					

Achieved

LEARNER ACTIVITY 7

♦ If a man runs 5 km in 30 minutes, at what speed is he running?

♦ Draw a set of axes and plot the following points:

⇒ H (1;2) , I (4;0) , J (3;-4) , K (-1;6) , L (-4;5) , M (0;3) , N (2;-2)

(Remember ! The x -axis is written first, then the y -axis)

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$= \frac{5 \text{ km}}{0,5 \text{ hrs}}$$

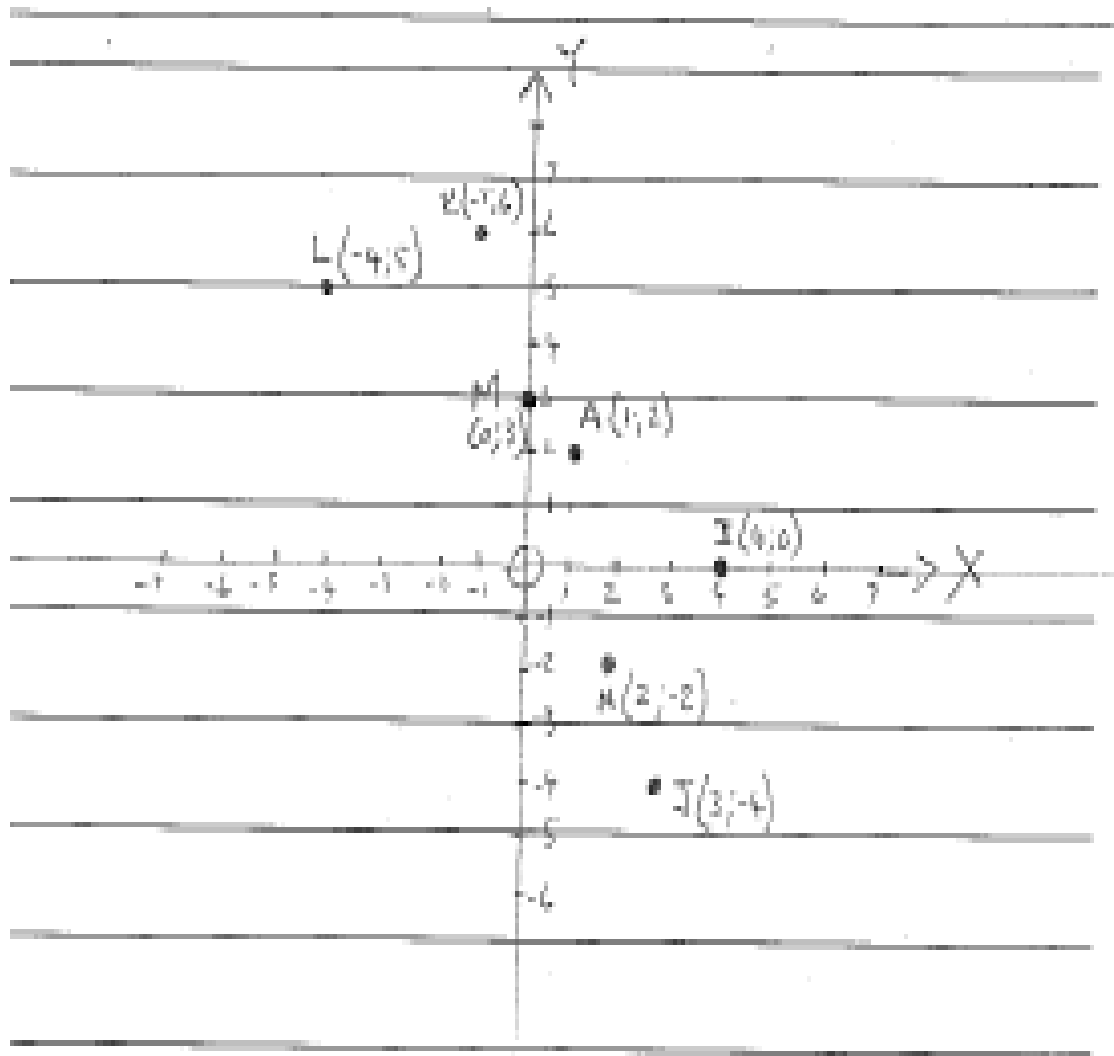
$$= 10\text{k/hour}$$

$$30\text{min} = 0,5\text{hrs}$$



LEARNER ACTIVITY 8

- ♦ On a plan the dimensions of a room are 35 mm × 45 mm. Give the actual dimensions of the room if the scale of the drawing is:
 - i) 1 : 1000
 - ii) 1 : 200
 - iii) 1 : 150
 - iv) 1 : 800
- ♦ Using a scale of 1 : 250, draw an accurate plan of a rectangular garden measuring 25 m × 15 m. Draw in a rectangular shed measuring 5 m × 4 m



Requirements achieved	YES	☒	☐	NO	☐
Comments in support of decision					

Achieved

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Explain what is meant with accuracy.			☒
2	Explain the term rounding.			☒
3	Do basic calculations using: addition, subtraction, multiplication, division, powers and roots.			☐
4	Apply the measurement system and instruments used in South Africa as set out in this guide.			☒
5	Apply are and volume calculation within an operational environment.			☒
6	Compile a rough sketch of an area or scene of incident.			☐
7	Interpret a plan drawn according to scale.			☐



Evidence Collector Signature

26.06.2021

Date



Security Officer Signature

22.06.2021

Date

Learner workbook



